materials naturally occurring in a fractured condition; or materials naturally occurring of highly angular nature or rough texture.

(b) Fine Aggregate – Fine aggregate shall be clean, hard, durable, moderately sharp, and free from coatings of clay, silt or other deleterious material and shall contain no clay balls or other aggregations of fine material. All aggregate passing the 4.75 mm sieve shall be non-plastic when tested in accordance with ASTM D4318.

(c) Mineral Filler and Mineral Dust – Mineral filler and mineral dust shall be free from organic matter. Mineral filler shall be non-plastic.

508.12.06 Blinding Sand – The Contractor shall be responsible for the supply of blinding material meeting the requirements of Table 508-D. That portion of blinding sand aggregate passing the 4.75 mm sieve shall be non-plastic when tested in accordance with ASTM D4318.

Table 508-D: Gradation for Blinding Sand

Sieve Size (mm)	Percentage Passing
4.75	100
2.36	50 – 100
1.18	30 – 80
0.300	10 – 55
0.150	3 – 20
0.075	0 – 3

508.13 Graded Aggregate Seal Design – The contractor shall provide a graded aggregate seal coat design, professionally prepared by qualified and experienced personnel. For this purpose, the Contractor shall use a qualified registered member of the Association of Professional Engineers and Geoscientists of British Columbia or a qualified registered member of the Applied Science Technologists and Technicians of British Columbia.

This design shall include application rate of the asphalt emulsion, in litres per square metre, spray temperature and the aggregate spread rate in kilograms per square metre. The design shall be presented to the Ministry Representative for review, at least three (3) days prior to the commencement of work and at any time subsequent changes are made to these rates.

The Contractor shall submit to the Ministry Representative four litres of a representative sample of the emulsion(s) and a twenty-kilogram sample of the aggregate(s) intended for use on the project. These samples shall be supplied at least three (3) days prior to the commencement of the work.

EQUIPMENT

508.20 General – The Contractor shall supply plant and equipment necessary for the proper and efficient completion of the work, including clear and effective means of calibration.

508.20.01 Access – The Ministry Representative shall have access at all times to any plant, equipment or machinery to be used on the Project.

508.20.02 Adjustment – If plant or equipment adjustment is necessary, it shall be done by a competent operator or mechanic or instrument repair technician, as required.

508.21 Equipment Characteristics – The following SS 508.21 provisions are the Ministry's preferred equipment characteristics but are not mandatory.

508.21.01 Pressure Distributor – The pressure distributor used for applying asphalt materials is to consist of a fully insulated tank with a minimum capacity of 5,000 L, permanently and rigidly mounted on a truck or fifth wheel trailer provided with pneumatic tires and propelled by a power unit capable of maintaining a constant and uniform speed.

The pressure distributor should be provided with the following equipment:

(a) A quick opening gate in the dome.

(b) A dipstick calibrated in 20 mm increments and a calibration chart showing the volume for each 20 mm of depth or provided the manufacturer dipstick and chart. Readily visible external depth gauge indicating the quantity of asphalt in the tank to the nearest 50 L.

(c) A means of heating the asphalt material uniformly to any temperature up to 125°C and maintaining the contents constantly at any selected temperature without any local overheating.

(d) A heavy duty, industrial-type thermometer and a remote reading dial thermometer.

(e) Rear-mounted spray bars and nozzles set parallel to the surface to be sprayed and capable of vertical and lateral adjustment. The spray bars to be capable of adjustment to provide a uniformly sprayed surface from a minimum of 0.6 m to at least 4 m width.

(f) The spray bars to be capable of producing up to triple coverage at one pass. The feed to the spray bars must be arranged so that there is a constant circulation of hot asphalt material to the bars to ensure a uniform viscosity and constant pressure of the liquid asphalt at each nozzle both before and during spraying operations.

(g) The feed manifolds to the spray bars are to be provided with positive filtration to prevent clogging of bars and nozzles.

(h) Rotary adjustable spray bar nozzles to be of the same type, correct size and manufacture; and set to produce uniformly fan-shaped sprays without atomization. Nozzles to be inspected daily and any nozzle having nicked, or damaged edges should be replaced.

(i) Spray nozzles to be provided with valves capable of instant full opening and positive cut-off.

(j) A connection to a circulating pressure manifold to which a hose may be attached for a single nozzle hand-operated outlet.

(k) A pressure pump capable of applying the asphalt material at a minimum rate of 180 L per minute, per metre of spray bar.

(l) A pressure gauge accurate to within 15 kPa and a meter calibrated in litres per minute should be readily visible.

(m) A fifth-wheel tachometer, calibrated in metres per minute and readily visible to the operator of the distributor.

(n) Vehicle spring "tie-downs" to prevent the change in height of the spray bars due to the reduction of the asphalt load during spraying.

(o) Electronic distance measuring device, with print out connected to the transmission, that is capable of measuring each spray length and the total length sprayed during the day.

508.21.02 Mechanical Spreader – Cover aggregate to be spread by means of an efficient, self-propelled spreader mounted on pneumatic tires and capable of continuously and uniformly spreading closely regulated quantities of crushed aggregates at the application rates selected. The loading hopper should be of such a capacity as to ensure continuous operation between dumps of aggregate. Aggregate to be transferred from the loading hopper to the front spreading device in such a manner as to ensure a uniform flow and prevent segregation of particles.

The spreading mechanism to be so designed as to distribute and spread aggregate across the selected width without segregation. Aggregate to be applied to the freshly sprayed surface by use of a comb screen for Class A and B Graded Aggregate Seals so that the largest particles are first placed on the road with the smaller fractions falling on top. The comb screen is to be mounted beneath the discharge openings of the spreader mechanism in such a manner that the angle of the comb screen relative to the road surface can be easily and quickly changed. Reverse flow aggregate spreaders are exempt from this clause.

Suitable provision to be made to prevent aggregate from rolling on the freshly sprayed surface. The cover aggregate should be applied ahead of the spreader wheels, with sharply defined, straight edges and without any lateral movement of the aggregate after initial application.

The spreader is to be designed to apply aggregate at any application width between 2 m and 4.12 m. A 4.26 m hopper is required.

508.21.03 Rollers – Self-propelled pneumatic tired rollers (minimum mass of 11,000 kg) to be used on the work.

All rollers shall be provided with equipment for spraying water continuously on all tires or wheels, while rolling is in progress, and be provided with such means as may be required to keep the tires clean and free of adhering asphaltic binder.

508.21.04 Power Brooms – Brooming work to be carried out using rotary power brooms. The brooms are to have sufficient width, power and brushing capacity to completely clean the surface of the standard lane to be treated or which has been treated within three coverages.

508.21.05 Water Truck – Water trucks to have a minimum capacity of 9,000 L. Water to be applied from a distributor of the pressure type, equipped with spray bar mounting nozzles similar to those used on asphalt distributors and capable of applying the water accurately and uniformly.

Splash plate type of distributors or those equipped with spray bars that eject fine streams of water will not be permitted.

The distributor to be provided with a satisfactory means for accurately measuring the quantity of water sprayed. Measuring equipment is to be calibrated. The Contractor shall make all necessary arrangements for obtaining water.

508.21.06 Tanks, Mobile Asphalt Heating, & Storage – Insulated liquid asphalt storage tanks should meet the following minimum criteria:

(a) A heating system, capable to heat from 10°C to 200°C.

(b) An automated heating and control system with safety shutdown.

CONSTRUCTION

508.31 Daylight Hours Only – Construction shall be carried out during daylight hours only.

508.32 Traffic Control and Signing – The Contractor shall submit, for the approval of the Ministry Representative, a Traffic Control Plan at least seven (7) days prior to the pre-construction meeting.

Traffic Control on Seal Coat projects shall be in accordance with SS 194 as well as the Ministry's publication "Traffic Control Manual for Work on Roadways".

The Traffic Control shall be maintained at all times when in the opinion of the Ministry Representative, the risk of damage to the mat is likely to become a safety issue or to result in unnecessary vehicular damage from loosened rock. Additional traffic control to protect the quality of the product is also the responsibility of the Contractor.

Traffic through the work zone shall be limited to 50 km per hour, or as directed by the District Manager Transportation, until completion of final sweeping.

Extending traffic control beyond the end of the shift, in order to maintain safety and reduce vehicular damage risks, will not preclude the Contractor from providing and applying blinding material to reduce these risks.

The Contractor shall continue to maintain flag-person assisted traffic control over the extent of the day's production, for a minimum of 24 hours for work on numbered routes, 2 hours on other routes, or as specified in the Special Provisions.

508.33 Temperature and Weather – Air temperature shall be a minimum of 6°C for application of graded aggregate seals, and application shall cease when the temperature falls below that level.

No application shall take place when the weather is misty or rainy.

508.34 Grade Preparation – Immediately prior to the application of the graded aggregate seal coat on a previously constructed granular surface, the surface shall be watered, graded and compacted and/or broomed to provide a non-segregated, tightly knit surface free from loose rock. Cross-fall will be at 3% or as directed by the Ministry Representative and super-elevations developed uniformly and carried consistently through each curve.

508.35 Joints – All joints between adjacent passes of seal shall be at centreline or lane lines.

508.36 Communications – The flag people, pilot car or cars, sign truck, spreader and supervisor's vehicle shall have an effective electronic means of communication to relay instructions and information accurately and quickly over the total length of the control area.

508.37 Sweeping – Sweeping shall meet the requirements of SS 145, SS 165, and the following:

(a) Immediately prior to the commencement of the sealing operations, all loose aggregates, dust, dirt, caked clay or foreign materials shall be removed from the width of the surface to be treated by brushing with power brooms supplemented by hand push brooms, shovels or the use of a power grader, and where necessary by flushing.

(b) Particular care shall be taken to thoroughly clean to outside edges of the strips to be treated and to ensure that the sweepings are not so deposited on the shoulder as to permit subsequent contamination of the treated surface.

(c) Following completion of sealing operations. the Contractor shall remove all loose aggregate remaining on all surface treated and paved surfaces to the satisfaction of the Ministry Representative. The Contractor will provide a minimum of 72 hours' notice of when the removal of loose aggregate will be complete and ready for inspection. The Ministry Representative will provide written acceptance that the removal of loose aggregate from all surface treated and paved areas has been satisfactorily been completed.

508.38 Repair – Any patching or repair work required, in the opinion of the Ministry Representative, before sealing operations begin, may be done by the Contractor as directed by the Ministry Representative and paid for under Extra Work.

508.39 Double Seal, Second Pass – Where the design calls for a double seal, the second pass of the surface treatment shall commence not less than 24 hours after completion of the first pass, subject to acceptable weather conditions as detailed in SS 508.33, and as the Ministry Representative may direct.

QUALITY CONTROL AND QUALITY ASSURANCE

508.41 Quality Control by the Contractor – The Contractor shall be responsible for providing all resources required to carry out the quality control on all the constituent materials, processes and products, their testing and inspection, within the construction of the seal coat and the quality of the end product (See SS 508 APPENDIX 508-E – Quality Control Requirements and Guidelines).

508.41.01 Quality Control Plan – The Contractor shall prepare and submit a Quality Control Plan, in accordance with the Contract requirements, at least seven (7) days prior to the pre-construction meeting.

508.41.02 Sampling, Testing and Observing – The contractor shall provide properly maintained equipment, qualified personnel and shall maintain access to a qualified laboratory so that quality control tests and observations are performed and the results must be reported to the Ministry Representative within 48 hours. For asphalt emulsions, the test results must be received within 21 days of sampling.

The Contractor's Quality Control Plan shall provide details of the equipment provided, its calibration, and application methodology to meet the Contract requirements.

508.42 Quality and Quantity Control Records

508.42.01 Emulsion Quality – For each batch of emulsion supplied to the job Site, the Contractor shall obtain from the supplier certified results for each test specified in Table 508-E (APPENDIX 508-A), and provide those results to the Ministry Representative prior to incorporating the emulsion into the Work.

When certified batch results are not yet available, the Contractor may elect, at its own risk, to use the uncertified material. If the material is subsequently determined to be non-compliant with the specifications or certified batch

results are not provided, remediation of any Work using such material will be to the Contractor's account.

508.42.02 Aggregate Quality– The Contractor shall assess the supplied aggregate to determine and ensure the quantity and quality for the intended application. Any modifications deemed necessary to ensure a quality product shall be made at the Contractor's expense.

508.42.03 Emulsion Usage – Daily emulsion usage shall be made available to the Ministry Representative on the Form "Emulsion Usage" (APPENDIX 508-C). Information shall be made available to the Ministry Representative within 48 hours and prior to demobilization of the storage tanks and pressure distributor from the Site.

Emulsion usage will be reconciled against initial tank volume, delivery records and remaining volume prior to the Contractor demobilizing from the Site.

508.42.04 Aggregate Usage – The Contractor shall monitor the aggregate spread rate on a production day basis and make the results available, in an acceptable format, to the Ministry Representative within 48 hours.

508.42.05 Daily Activity Log – The Contractors' Superintendent or designate shall complete and make available to the Ministry Representative the "Daily Activity Log" (APPENDIX 508-D) within 48 hours. This form shall be completed and submitted for all calendar days from the first day of mobilization to the last day of demobilization, for any day on which work was performed on-site.

508.42.06 Emulsion Sampling

(a) The Contractor shall, procure and store a four litre sample of the emulsion from each tanker for use by the Ministry and/or Contractor.

(b) These samples shall be obtained, handled and stored in accordance with ASTM D140 and also with the manufacturer's recommendations.

(c) The samples shall be identified and retained by the Contractor until demobilization at which time permission may be granted by the Ministry Representative to discard the samples in an environmentally friendly manner.

(d) The Ministry Representative may provide the Contractor with a copy of the test results from the Ministry's Quality Management program. The Contractor shall not utilize these tests in place of their own Quality Control Program.

(e) The absence of Ministry test results shall not relieve the Contractor's obligation to remedy any defect in materials or construction.

508.42.07 Final Quantity Report – The Contractor shall supply to the Ministry Representative, prior to final acceptance of the work, the total amount of emulsion and graded aggregate seal aggregate used on the project, and the volume of unused emulsion remaining in the tank(s).

END PRODUCT ASSESSMENT

508.51 General – The Contractor shall provide a product conforming in accuracy and detail as specified by the contract. Payment is based on end product assessment by the Ministry Representative.

508.52 Surface Deficiency Definitions – See APPENDIX 508-F for examples of surface conditions, photos and details of the conditions listed below.

(a) **Blackening –** Blackening is the emergence of excess asphalt onto the finished surface whereby the finished surface becomes discoloured, but the coarse surface texture remains. Under this condition, the effective cover stone can still be seen to protrude above the level of the asphalt binder's upward migration.

(b) **Flushing and Bleeding –** Flushing and bleeding is the emergence of excess asphalt and fines onto the finished surface and the significant loss of coarse surface texture, leaving little or no effective cover aggregate visible.

(c) **Pock Marks –** Pock Marks are indentations in the mat texture, normally less than the thickness of the graded seal layer, evidenced by the absence of effective cover aggregate in a small area (No photograph available).

(D) **Potholes –** Potholes are the loss of all cover aggregate and base material(s) into the underlying asphalt or gravel base. Potholes may develop because of seal and/or base weakness.

(e) **Raveling –** Raveling is a general term for loss of cover aggregate, which can include numerous pock marks.

(f) **Streaking –** Streaking is a continuous visible absence of effective cover aggregate often seen paralleling a 'ridge' of excess cover aggregate, characterized by the narrowness of its width and the significant extent of its length. Where streaking displays an absence of effective cover aggregate, the condition is unlike raveling in that there was originally little or no effective aggregate in the area.

(g) **Washboard –** Washboard is a condition of non-uniform surface texture caused by the buildup of transverse ridges within the graded seal layer itself, and characterized by a very rough, vibratory or chattering ride (No photograph available).

508.53 End Product Evaluation

508.53.01 Lot Definition – A Lot shall be one lane kilometre of highway sealed**.** Sections of any road/highway remaining which are less than one-half (0.5) kilometres in length, will be included in the previous Lot. Sections remaining which are 0.5 kilometres or greater, but less than one kilometre will receive a Lot designation unique for the road/highway.

508.53.02 Rating Time Criteria – The Ministry Representative will rate each Lot, within thirty (30) days of final brooming.

508.53.03 Performance Rating Criteria – Performance rating will be in accordance with APPENDIX 508-F, Table 508-G, Table 508-H, and the photographs.

508.54 Deficiency Repair and Replacement – Any Lot of sealed surface that is found unacceptable shall be remedied in accordance with Table 508-G (APPENDIX 508-F), until the whole project has been repaired or replaced as necessary and accepted by the Ministry Representative.

On double seal applications both the lower and upper lifts must be free of deficiencies prior to being accepted.

The sealing Contractor will not be responsible for deficiencies attributed to base, base failures and base work done by others. These may be repaired under a Provisional Sum Item provided for that purpose, or as ordered by the Ministry Representative.

For any defects being repaired through re-seal, the repair must cover the entire width of the highway (or to a centerline barrier), unless the Contractor can demonstrate that a compliant repair can be achieved at a narrower width without any longitudinal ridging which may adversely affect driver control or trap water adjacent to the edge of the repair.

508.55 Attention to Potential Safety Hazards – If, in the opinion of the Ministry Representative, remedial Work is required to correct a potential safety hazard it shall be undertaken immediately.

MEASUREMENT AND PAYMENT

508.61 Surface Seals – Surface Seal will be measured and paid by the square metre.

Payment in a double pass application will be for final top lift surface area only.

Payment will be full compensation for pre-cleaning the surface; loading, hauling, placing the asphalt emulsion and aggregate; compaction; brooming the finished surface; quality control and traffic control.

No extra payment will be made for any additional equipment, tools, labour or material used to replace, repair or overlay Unacceptable Work.

No extra payment will be made for any additional equipment, tools or labour employed to satisfy special brooming requirements.

508.62 Asphalt Emulsion – Asphalt emulsion will be measured and paid in litres of emulsion actually sprayed on the road, including supply, storage, heating and handling.

508.63 Aggregates – Aggregates will be measured by the cubic metre in stockpile. The volume will be determined by surveying and surface-to-surface volume calculations. Measurement shall be performed by the Contractor at the Contractor's expense using an independent surveying company. Survey results must be verified and signed off by the independent surveying company and survey results must be supplied to the Ministry.

Payment for aggregates will be at the Contract Unit Price per cubic metre in stockpile. Payment for aggregate will be only up to 100% of the Approximate Quantity as stated in Schedule 7.

No payment will be made for the supplying, hauling and placing of blinding sand.

The supply of extra aggregate, required to repair or overlay Unacceptable Work, will be the Contractor's responsibility. Where sufficient aggregate quantities are remaining in stockpile, the Ministry will make these available at no cost to the Contractor.

508.64 Stockpile Bases – The stockpile base must be prepared prior to surveying and stockpiling. No payment will be made for stockpile base preparation work and no payment will be made for materials required for the preparation of stockpile bases.

508.65 Claims for Vehicular Damage – Without in any way limiting the Contractor's obligations or liabilities herein, during construction and up until final acceptance of the whole project by the Ministry Representative, the Contractor shall be fully responsible for all claims for damages caused by the work and shall address and deal with each claim submitted.

508.66 Partial Payment for Rejected Work – In the Ministry Representative's sole discretion and without setting precedence, where any work is reject but the Ministry Representative determines that it may be left in place, the Ministry Representative may authorize partial payment to the Contractor as full compensation for any residual value the work may have. Notwithstanding the foregoing, the Ministry is under no obligation to make any payment for reject work.

APPENDIX 508-A

QUALITY CONTROL TESTS FOR HIGH FLOAT EMULSIONS

Quality Control Tests, to be obtained by the Contractor from the Supplier, for High Float Emulsified Asphalts.

Table 508-E: Quality Control Tests for High Float Emulsified Asphalts

Designation	ASTM Test Title
D5	Standard Test Method for Penetration of Bituminous Materials
D88	Standard Test Method for Saybolt Viscosity
D113	Standard Test Method for Ductility of Asphalt Materials [Ductility Test for Polymer Modified Emulsions]
D139	Standard Test Method for Float Test for Bituminous Materials
D140	Standard Practice for Sampling Asphalt Materials
D244	Standard Test Methods and Practices for Emulsified Asphalts [Residue by Distillation]
D4957	Standard Test Method for Apparent Viscosity of Asphalt Emulsion Residues and Non-Newtonian Asphalts by Vacuum Capillary Viscometer [Absolute Viscosity of Asphalts]

APPENDIX 508-B

HIGH FLOAT EMULSIFIED ASPHALT REPORT (H-539)

Project No:_____ **Refinery:** _____

Contract:_____ **Asphalt Grade:** _____

Location:_____ **Batch No:** _____

Sampling Date:_____ **Testing Date:** _____

Requirements	Results	Specifications	
		Maximum	Minimum
Residue by Distillation, % by mass			
Oil Distillation, % by volume			
Saybolt Furol Viscosity (SFS), 50°C			
Penetration, 5 s/100 g/25°C			
Viscosity, mK or CM (Pa•s), at 60°C			
Ductility, at 4°C			
Float Test (s), at 60°C			

APPENDIX 508-C

EMULSION USAGE REPORT

Project No.: _____ **Product:** _____

Date: _____ **Supplier:** _____

Contractor: _____ **Supplier Lading No.** _____

Emulsion Temp. (°C)	EMULSION QUANTITIES (litres)			SPRAY COVERAGE (m or m^2)			APPL. RATE (L/m^2) [L/F]	LOCATION† (Sta. to Sta. or LKI, and lane being sealed)
	Start [A]	Finish [B]	Used [L=A-B]	Width [D]	Length [E]	Area [F=DxE]		

TOTALS: _____ _____ = Sums for "Used" and "Area"

AVERAGE: _____ = Total Used / Total Area

The Undersigned certifies that the quantities placed are as indicated in the information above.

Name (Contractor Operator): _____ **Date:** _____

Checked by (Ministry Representative): _____ **Date:** _____

† **Example location description:** *Sta. 3+500 to 5+100, WBSL* (where WBSL = Westbound slow lane)

APPENDIX 508-D

DAILY ACTIVITY LOG

Location:		**Emulsion Supplier:**	
		Emulsion Product:	
Contractor:		**Aggregate Source:**	
Superintendent:		**Aggregate Class:**	

(Note: Enter all time in a 24 hour clock format, e.g 3:00 PM → 15:00)

Equipment

Type	Unit #	Make	Year

Aggregate Haul:	Startup	Shutdown
# of Haul Units:		
Total Loads Today:		

Weather	Startup	Shutdown
Air Temp:		
Road Surface Temp:		

Traffic Control	Startup	Shutdown
Shift 1 (Morning) Time:		
# of Pilot Vehicles:		
# of TCP's:		
Shift 2 (Afternoon) Time:		
# of Pilot Vehicles:		
# of TCP's:		
Shift 3 (Night) Time:		
# of Pilot Vehicles:		
# of TCP's:		

Production

Start up Time:		Total m^2	
Shutdown Time:			

Emulsion:

Load #	Time In:	Litres	Invoice #	Notes

Total Litres In: Litres at Shutdown:

Delays:

Time		
From:	To:	Cause

Notes

Contractor Representative Signature: _____

Quality Control Manager Signature: _____

APPENDIX 508-E

QUALITY CONTROL REQUIREMENTS AND GUIDELINES

508-E.01 GENERAL – The Contractor's guidelines for preparing, submitting and adhering to a Quality Control Plan are specified in SS 101, SS 508, the Special Provisions and this APPENDIX 508-E.

The QC Manager shall have at a minimum five (5) years related experience in Quality Management.

508-E.02 QUALITY CONTROL PLAN – The Plan shall address all elements that affect the quality of the materials, end product seal coat, including but not limited to the following:

(a) Supply of Asphalt Materials

(b) Identification of source and proof of quality of aggregates to be supplied

(c) Aggregates production and its gradation control

(d) Quality of aggregate components during production

(e) Stockpile management

(f) Sealcoat Design(s)

(g) Distributor and spreader calibration

(h) Process temperature controls

(i) Material application rates

(j) Joints and tapers

508-E.03 CONTRACTOR'S RECORD OF QUALITY CONTROL TESTING – Test results should be made on specified forms or charts immediately after completion of each test. These test results are to be made available to the Ministry Representative upon request.

Records of gradation control during aggregate production should be kept on the form H-295 – Mechanical Analysis of Aggregates (available from the Ministry Representative) or on a similar gradation analysis form acceptable to the Ministry Representative.

508-E.04 MATERIAL APPLICATION RATE –The Contractor shall control the Material Application Rate by monitoring the amount of emulsion and aggregate delivered to the road against the area covered by checking the application rates of the distributors and spreader.

508-E.05 OTHER QUALITY CONTROL PROCEDURES – The Contractor may initiate other Quality Control procedures as necessary for ensuring production of a quality product and include them in a Quality Control Plan. Procedures may also be introduced after the start of work as amendments to the Quality Control Plan.

508-E.06 QUALITY CONTROL TESTING FREQUENCY – Recommended test frequency requirements for Quality Control are described in Table 508-F.

508-E.07 QUALITY ASSURANCE TESTING FREQUENCY – Quality Assurance Testing shall be approximately 10% of that described for Quality Control and may vary in accordance with the Quality Control Plan and the observed effectiveness of Quality Control.

Table 508-F: Test Frequency Guidelines

Activity	ASTM Test Designation	*Minimum Frequency
Tests Before Aggregate Production	D75, Standard Practice for Sampling Aggregates. Sampling Stone, slag, gravel, sand and stone block for use as Highway Materials	When Required
	C117, Test for Materials Finer than 75 µm (No.200) Sieve in Mineral Aggregates by Washing	One (1) test (Representative Sample)
	C142, Test Method for Clay Lumps and Friable Particles in Aggregates	One (1) test (Representative Sample)
	D6928, Micro Deval	One (1) test (Representative Sample)
	D5821, Fracture Count on Coarse Aggregate	One (1) test (Representative Sample)
	D4791, Test Method for Flat Particles, or Flat and Elongated Particles in Coarse Aggregate	One (1) test (Representative Sample)
Tests During Aggregate Production	C136, Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates	Main Stockpile – One (1) per each 1.5 hours production.
	D5821, Determining the Percentage of Fractured Particles in Coarse Aggregate	Main Stockpile – One (1) per each 1.5 hours production.
	C117, Test for Materials Finer than 75 µm (No 200) Sieve in Mineral Aggregates by Washing	One (1) per shift on reduced sample obtained from combined samples from the crusher.
Tests on Emulsified Asphalt Products	D244, Residue by Distillation	Minimum one (1) per batch
	D244, Saybolt Furol Viscosity	Minimum one (1) per batch
	D5, Penetration of Bituminous Materials	Minimum one (1) per batch
	D139, Float Test	Minimum one (1) per batch
	D4957, Apparent Viscosity on Residue	Minimum one (1) per batch
	D113, Ductility	Minimum one (1) per batch on polymer modified emulsions.
Tests on Application of Sealcoat Materials	Emulsion Pan Test	Three (3) per distributor/day. Test procedure, variation limits and actions should be outlined in Contractor's Quality Control Plan.
	Aggregate Pan Test	Two (2) per spreader/day. Test procedure, variation limits and actions should be outlined in Contractor's Quality Control Plan.

* *These are the recommended minimum frequencies and the Contractor is responsible to assess the need to increase testing when required.*

APPENDIX 508-F

SURFACE DEFECTS

Table 508-G: Surface Defect and Treatment Repair / Replacement Criteria

Surface Defects	Severity	Rating Parameters	Repair/Replacement Method
Loss of cover aggregate (Raveling)	Very good	No noticeable aggregate loss	None
	Good	A few pock marks, less than 5 per 0.1 m^2	None
	Fair	Frequent pock marks closely spaced, more than 5 per 0.1 m^2	Reseal of affected area***
	Poor	Extensive pock marks or few surface disintegrations	Reseal of affected area***
	Very poor	Disintegration with potholes	Reseal of affected area***
Potholes	Very good	Few minor potholes, only involves graded aggregate seal	All potholes to be filled with asphaltic patching material at contractor's expense
	Good	Few deep potholes, involves granular base	All potholes to be filled with asphaltic patching material. (see SS 508.54)
	Fair	Intermittent potholes	Patch potholes and reseal affected area*** (see SS 508.54)
	Poor	Frequent potholes	Patch potholes and reseal affected area*** (see SS 508.54)
	Very poor	Extensive or potholes throughout	Patch potholes and reseal affected area*** (see SS 508.54)
Flushing / bleeding	Very good	No/very faint colour change in wheel path	None
	Good	Few sections with asphalt on surface	None
	Fair	Intermittent sections with asphalt on surface	Application of sand blotter
	Poor	Frequent sections with asphalt on surface, has wet look or asphalt on surface throughout	Reseal of affected area. Removal of initial surface treatment at Contractor's option***
	Very poor	Wet look with noise like a wet pavement	Reseal of affected area. Removal of initial surface treatment at Contractor's option***
Total failure	Any	Any condition where the asphalt material softens or disintegrates under traffic and aggregate is picked up or "kicked off" by traffic	Remove and dispose of failed surface treatment in its entirety and apply new surface treatment
Streaking (see raveling)	Any	Absence of cover aggregate caused by too little binder – characterized by long, narrow sections of little or no stone cover	Reseal of affected area
Washboard	Any	Any condition of repetitive build-up across the travel lane	Remove and reseal affected area

*NOTE: *** Reseal for these defects shall consist of a new graded aggregate surface treatment, single pass application. Area of the reseal repair/replacement shall not be less than one lane width, 10 metres in length. If there is less than 10 metres between two sections in the application pass designated for repair/replacement, the repair/replacement shall be continuous.*

Table 508-H: Surface Defects Frequency Definitions

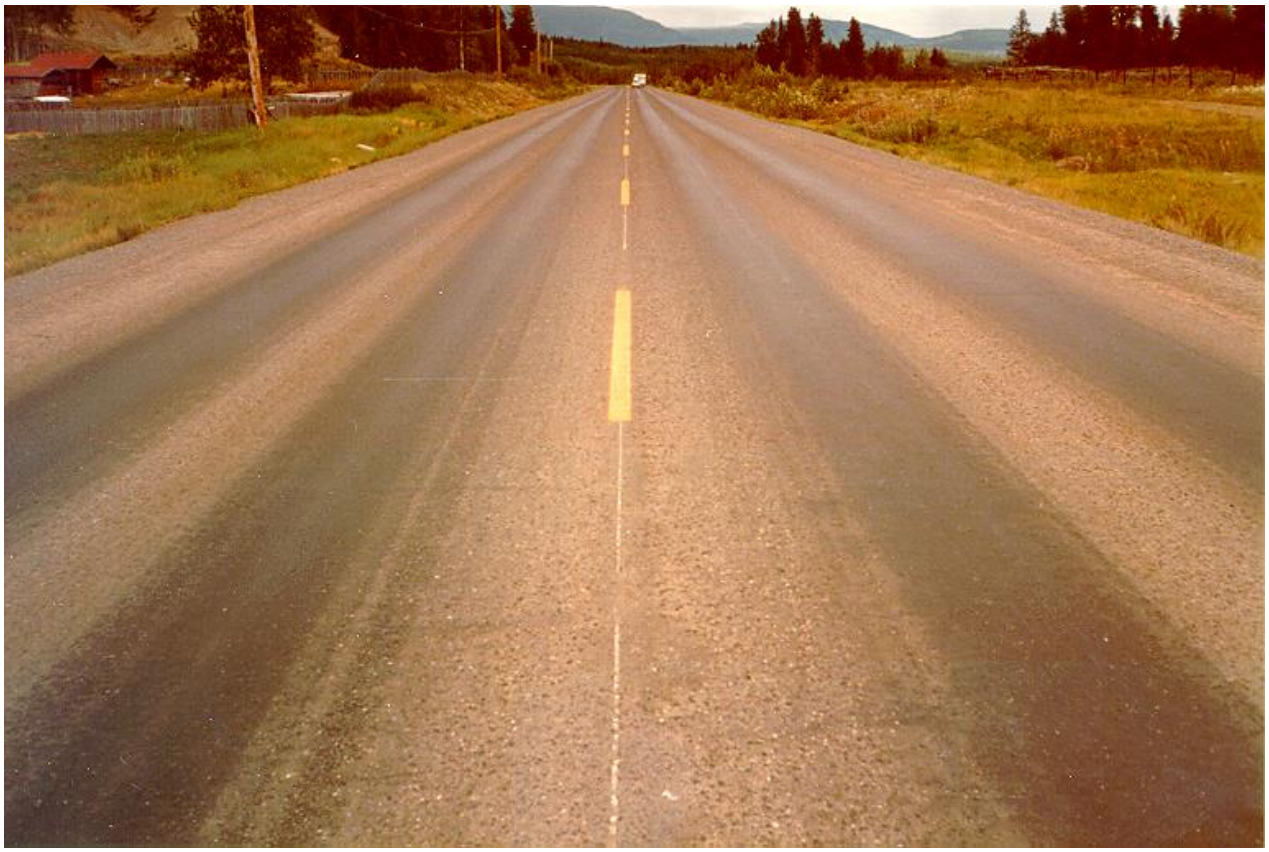
Density of Surface Defects**			
Severity	**Raveling (% Length)**	**Potholes (Number)**	**Flushing/Bleeding (% Length)**
Very Good	<5%	0 – 1	<5%
Good	5% – 20%	2 – 15	5% – 20%
Fair	21% – 50%	16 – 30	21% – 50%
Poor	51% – 80%	31 – 50	51% – 80%
Very Poor	>80%	>50	>80%

*Note:** Based on number of surface defects or percent of surface affected by defect, per Lot as defined in SS 508.53.01.*



Blackening





Flushing or Bleeding

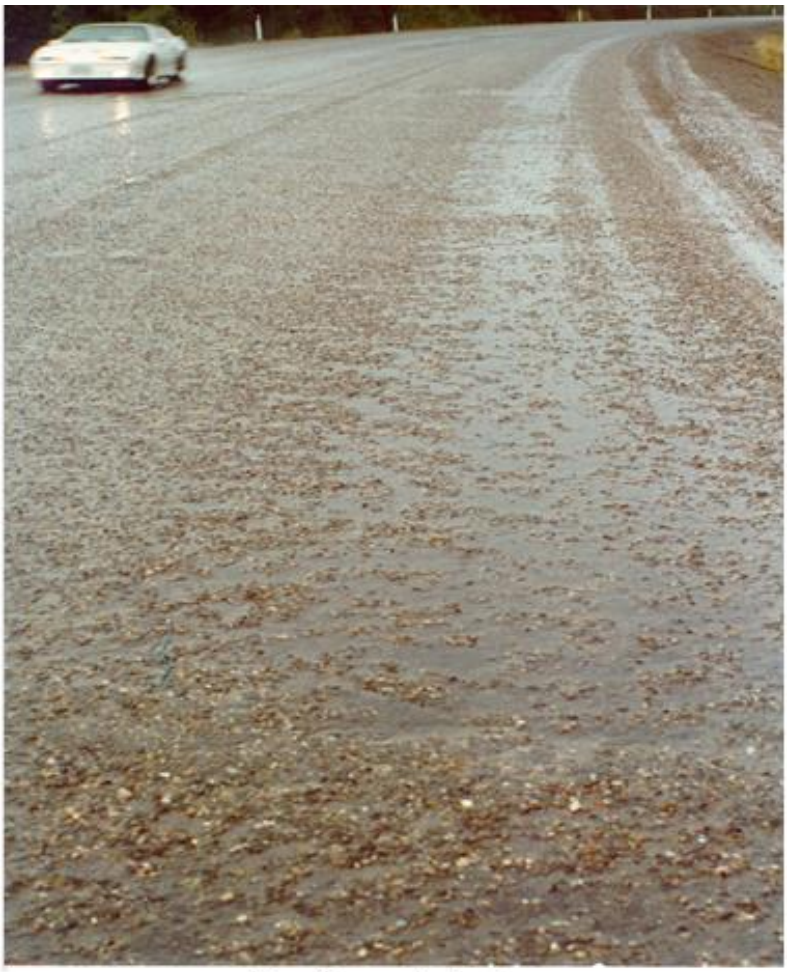




Pothole



Streaking



Raveling or Stripping



SECTION 511
COLD MILLING

DESCRIPTION

511.01 Scope – This work consists of preparing a foundation for placement of new surfacing by cold milling the existing pavement surface to the depth and width as indicated on the Drawings.

EQUIPMENT

511.21 Milling Machine – The milling machine shall be self-propelled and shall be equipped with automatic longitudinal and transverse grade control, which shall be used when directed by the Ministry Representative. A profiling ski or boom of a recommended minimum 4 m length, approved by the Ministry Representative, shall be used. No substitute equipment will be permitted. The cutting drum shall be a minimum of 1.8 m in width, totally enclosed and with replaceable cutting teeth. There shall be an effective means of removing the loosened material from the surface and for preventing dust from escaping into the air.

511.22 "Kill" Switch – All pavement milling machines shall be equipped with a "kill" switch installed or approved by the manufacturer of the milling machine. This "kill" switch shall be automatically actuated whenever the operation of the machine is so impeded that a hazardous situation, such as "kick back", would result were that operation to continue. When so actuated the "kill" switch shall instantly shut down the operation of the milling machine.

511.23 Audible Back Up Warning Device – See SS 145.30.04.

CONSTRUCTION

511.31 Running Surface for Lanes Open to Traffic – The Contractor shall ensure that a suitable running surface is maintained on any portion of the highway that is open to traffic during construction.

At all times, during the milling operation, the traveled roadway shall be kept clean of all loose materials.

511.32 Pavement Removal – The existing pavement shall be removed to the depth and width as specified in the Special Provisions, on the drawings or as directed by the Ministry Representative to provide a surface that is free of longitudinal and transverse irregularities. Milling may be specified as a nominal uniform depth, or as variable depth profile milling to achieve design lines and grades.

The use of a heating device to soften the pavement will not be permitted.

At the end of the milling shift or at the end of each section to be milled, prior to reopening the road to traffic, the pavement shall be cleaned and swept so that all loose material is removed. In addition, the transition from the milled surface to the existing paved surface shall be tapered at a maximum slope of 50H:1V, or as directed by the Ministry Representative. The Ministry Representative shall indicate whether the taper is to be milled or a tar paper letdown shall be constructed using asphalt mix.

When the milling operation traverses intersecting roads or on and off ramps, the transition from the milled surface to the existing pavement shall be at a maximum slope of 50h:1v or as directed by the Ministry Representative. This transition can be done with the milling machine or by using tar paper letdowns as approved by the Ministry Representative.

If, due to delays between the milling and paving operations, the milled surface starts to pothole or deteriorate, repairs shall be carried out at once using asphalt mix.

At those locations where milling is conducted, the Contractor shall cut drainage channels as required to prevent water from collecting in the milled area.

If washouts occur at any time during the milling operations, they shall be immediately repaired by the Contractor, at the Contractor's own expense, with approved materials and as directed by the Ministry Representative.

511.33 Disposal of Materials – The disposal of materials removed under this section, which are not recycled and used on the project, shall become the property of the Contractor at the point of removal, unless otherwise specified. Materials shall be disposed of outside the limits of the project in a manner satisfactory to the Ministry Representative and shall meet all environmental requirements. The procurement of a disposal site, if required, shall be the Contractor's responsibility.

511.34 Existing Structures – Care must be taken not to disturb or damage any structures or devices such as manholes, catch basins, valves, boxes and other utilities. All metal and concrete faces must be cleaned of old pavement.

The Contractor shall repair, at its own expense, any damage to visible, referenced, or plan-indicated manholes, catch basins, valves, valve covers and concrete/asphalt curb, or any other infrastructure.

MEASUREMENT

511.81 General – Cold Milling will be measured by the SQUARE METRE or the TONNE as specified in the

Contract. The area quantities will be determined by measurement of the actual surface area from which the materials have been removed and computed to the nearest square metres. Measurement of material by mass shall be done in accordance with SS 145.18.04 and SS 145.19 before moving to stockpile.

PAYMENT

511.91 General – Payment for COLD MILLING will be at the Unit Price bid per square metre or tonne as specified in the Contract. The unit price shall be for all things furnished and done including servicing all equipment, replacing teeth on the cutting drum, and performing all work to remove the existing pavement by milling, loading, hauling and placing the material in stockpile.

511.92 Asphalt Mix – Asphalt mix required to repair the milled surface as set out in SS 511.32 shall be paid at the Unit Price bid for top lift asphalt, except where, in the opinion of the Ministry Representative, the milled surface damage occurred during a delay that was the Contractor's responsibility, in which case there will be no payment.

SECTION 515

HOT-IN-PLACE RECYCLED ASPHALT PAVEMENT (EPS)

GENERAL

515.01 General – This Section describes the materials, equipment, professional standards, and end product requirements for the construction of Hot-In-Place Recycled Asphalt Pavement (HIP).

End Product Specifications (EPS) contain the acceptance and payment criteria based on the results of specified sampling and testing. Payment of the contract Unit Prices for the Hot-In-Place asphalt pavement product is subject to adjustments upward and downward in accordance to the provisions provided in EPS.

Standards applicable to this Section are listed in Table 515-A.

515.02 General Description of Work – The Contractor will be responsible for the following work associated with the construction of Hot-In-Place asphalt pavement by EPS:

(a) Preparing a Quality Control Plan for evaluation before commencing the Work and providing at the production site a testing facility to provide the data needed to implement that plan.

(b) Supplying, screening, crushing, processing and improving aggregate to produce virgin asphalt mix aggregate.

(c) Producing or supplying, and testing, all materials in accordance with Table 515-B through Table 515-E inclusive, and ensuring compliance with Contract specifications.

(d) Supplying and delivering asphalt cement meeting the requirement of SS 952.

(e) Rejuvenating Agent Requirements: The Contractor shall supply a rejuvenating agent from the Ministry's Recognized Product List and meeting the requirements shown in Table 515-A.

The Ministry's Recognized Product List can be found on-line at:

https://www2.gov.bc.ca/assets/gov/driving-and-transportation/transportation-infrastructure/engineering-standards-and-guidelines/recognized-products-list/recognized_products_list.pdf?forcedownload=true

Rejuvenating agents that contain water will not be permitted.

(f) Heating the virgin asphalt mix aggregate and mixing it with asphalt cement to produce asphalt admix that meets the Job Mix Formula.

(g) Hauling, and adding the asphalt admix into the recycling process.

(h) Recycling the existing highway pavement by heating, milling, reprocessing, with or without the addition of rejuvenating agents or admix (virgin mix) as specified, re-mixing, replacing and compacting using a multi-stage, hot milling process and rollers.

Table 515-A: Specifications

ASTM Test	Title of Test	Specification
D2726	Bulk Relative Density of Compacted Bituminous Mixtures Using Saturated Surface Dry Method	
D1188	Bulk Relative Density of Compacted Mixes Using Paraffin Coated Method	
D1559	Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus	
D2170	Kinematic Viscosity @ 60°C	200 – 500 mm^2/s
D70	Specific Gravity	0.98 – 1.02
D92	Flash Point by Cleveland Open Cup	204°C (Min)
D116 (10 mm Hg)	Volatility – IPB • 2% v. • 6% v.	150°C (min) 190°C (min) 210°C (min)
D2007	Saturates, %w	28 (max)
D2006 – 70	Asphaltenes, %w	1.5 (max)
D2006 – 70	Chemical Composition $(PC+A_1)/(S+A_2)$	0.4 – 0.8

Table 515-B: Standard Sampling and Testing Procedures for Virgin Aggregates

Item	Procedure	ASTM Test Reference
1	Sieve Analysis of Fine and Coarse Aggregates	C136
2	Materials Finer Than 0.075 mm (No. 200) Sieve in Mineral Aggregate by Washing	C117
3	Specific Gravity and Absorption of Coarse Aggregate	C127
4	Specific Gravity and Absorption of Fine Aggregate	C128
5	Determining the Percentage of Fractured Particles in Coarse Aggregate	D5821
6	Total Moisture Content of Aggregate by Drying	C566

All virgin aggregate tests shall use the following sieve sizes:

37.5 mm, 25 mm, 19 mm, 16 mm, 12.5 mm, 9.5 mm, 4.75 mm, 2.36 mm, 1.18 mm, 0.600 mm, 0.300 mm, 0.150 mm, and 0.075 mm.

Table 515-C: Standard Sampling and Testing Procedures for Virgin Asphalt Admix and Hot-In-Place Recycled Pavement

Item	Procedure	ASTM Test Reference
1	Bulk Specific Gravity and Density of Non-Absorptive Compacted Bituminous Mixtures	D2726
2	Theoretical Maximum Specific Gravity and Density of Bituminous Paving Mixtures	D2041
3	Sampling Compacted Bituminous Mixtures for Laboratory Testing	D5361
4	Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus	D1559 or AASHTO T 245
5	Determining Asphalt Content by Ignition Methods for Admix	D6307

Table 515-D: Standard Sampling and Testing Procedures for Asphalt Cement

Item	Procedure	ASTM Test Reference
1	Penetration of Bituminous Materials	D5
2	Viscosity of Asphalt by Vacuum Capillary Viscometer	D2171

Table 515-E: Test Sample Source Locations

Test	Sample Source
Density	150 mm diameter road cores
Smoothness	Centre of Lane

(i) Heating shall be regulated so no overheating of the existing asphalt cement will occur. This will generally require the existing surface to be radiantly heated and no open flame directed at the pavement will be permitted.

(j) The cutting drum(s) shall mill to the specified cross section and shall be set to produce a fully homogenous recycled mixture free of lumps.

(k) The activated heated screed shall be complete with augers and strike-off device in accordance with SS 502 and be capable of distributing and placing the reprocessed mix to the full width of the pavement being recycled.

Production shall have a minimum rate of 500 m^2 per hour except between June 1st and September 8th when the minimum rate shall be 700 m^2 per hour.

(l) Hot recycling unit shall be equipped, capable to add and homogeneously blend any specified proportion of virgin Asphalt Mix into the reclaimed material and shall be equipped with a means to introduce a rejuvenating agent homogeneously into the reclaimed mixture.

(m) Individual pre-heating and reprocessing units shall be equipped in such a way that the operator and other workers are not exposed to any hazardous fumes or gases produced from the equipment operation, the heated pavement or from the addition of the rejuvenating agent, in accordance with all applicable regulations.

The Contractor shall provide an end product conforming in quality and accuracy of detail to the dimensional and tolerance requirements of the contract. Where no tolerances are specified, the standard of workmanship shall be in accordance with normally accepted good practice.

Payment is subject to adjustments based on quality assurance tests performed by the Ministry Representative.

The Ministry shall provide information pertaining to the gradation of the admix, asphalt content, addition rates for rejuvenating agents, and the percentage of admix to be incorporated.

515.03 Definitions Relevant to End Product Specification

(a) **End Product Specification (EPS) –** A specification whereby the Contractor is entirely responsible for quality control of the construction processes, and whereby the Ministry performs the specified quality assurance sampling and testing of the end product for the purpose of determining acceptance/rejection and payment.

(b) **Job Mix Formula** – Establishes the virgin aggregate proportions, gradation, the asphalt content and type of asphalt to be used for production of asphalt admix. The Job Mix Formula is based on an asphalt mix design, in accordance with SS 515.05.04, or on a variation in accordance with SS 515.05.07.

(c) **Asphalt Admix** – Virgin asphalt mix added to the recycled mix. The Asphalt Admix is project specific designed and is blended homogeneously within the recycling process. The Asphalt Admix becomes an integral component of the recycled pavement structure.

(d) **Asphalt Content** –The quantity of asphalt cement in the virgin admix expressed as a percentage by weight of the total dry aggregate in the mix.

(e) **Design Asphalt Content** – The asphalt content upon which the Job Mix Formula is initially established.

(f) **Approved Asphalt Content** – The design asphalt content or subsequent adjustment to it, embodied in a Job Mix Formula or revised Job Mix Formula.

(g) **Actual Asphalt Content** – The amount of asphalt cement in the admix as determined by the Contractor's Quality Control Testing.

(h) **Smoothness** – A measure of the longitudinal profile of the pavement surface. The unit for measurement is the International Roughness Index (IRI).

(i) **Sample Mean** – The arithmetic mean of a set of test results constituting the sample.

(j) **Stratified Random Sample** – A set of test measurements taken one each from a number of separate (stratified) areas or Sub-Lots within a Lot in an unbiased way.

(k) **Lot** – A portion of the work being considered for acceptance and for the determination of payment. A Lot is defined as follows for the application of the contract requirements for:

(i) **Density**: A Lot shall be one day's scheduled production of at least 7 hours of pavement recycling where no changes have occurred to criteria such as but not limited to:

(A) Approved Job Mix Formula;

(B) Specific thickness to be processed;

(C) Required materials addition rate;

(D) Change in the ratio of the components making up the Hot-In-Place Recycled Pavement.

A change in any above criteria may require a new Lot designation.

One day's production of less than 7 hours will be dealt with as follows:

(E) The material will be added to the next Lot that has the same criteria, as described above, except that

(F) If a test indicates that this production is subject to a payment adjustment or to rejection, or if no further material will be produced with the same criteria, this production will be designated as a separate Lot.

A Lot shall be no more than two days total production even if the above criteria has not changed or been met.

(ii) **Smoothness**: A Lot shall be one kilometre length of recycled pavement for each driving lane.

(l) **Sub-Lots for Density and Smoothness** – For the application of the contract requirements of:

(i) **Density:** Each Lot shall be divided into three equal Sub-Lots.

(ii) **Smoothness:** Each Lot shall be divided into 100 metre Sub-Lots.

(m) **Driving Lane** – A single lane in any area of the pavement other than a shoulder or a barrier flare.

(n) **Surplus Aggregate** – Aggregate surplus to the Works, in split or un-split stockpiles which singly or combined will meet the virgin asphalt mix aggregate mix gradation. Surplus aggregate will be paid for as indicated in SS 515.32.01 There will be no payment for reject aggregate.

(o) **Hot-In-Place Recycled Pavement** – Shall be the results of the process where recycling, the addition of admix and/or rejuvenating agents, the spreading, compacting and finishing of the asphalt pavement have taken place to form a quality finished product.

The Hot-In-Place Recycled Pavement may be comprised of the following components: existing asphalt pavement, admix, and/or rejuvenating agents.

515.04 Quality Control

515.04.01 General – Quality Control, by the Contractor is the sum total of activities by the Contractor to ensure that a product meets contract specification requirements. Quality Control includes material handling and construction procedures, calibration and maintenance of equipment, production process control and any sampling, testing and inspection that is done for these purposes.

The Contractor is responsible for all Quality Control under this Section. The Ministry Representative will audit and monitor the Contractor's operations and the implementation of the Contractor's Quality Control Plan.

The Ministry will not take samples for quality control testing and will in no manner assist in any degree or in any aspect of the Contractor's operation in the production of asphalt pavement, beginning from the production of aggregate through the compaction of asphalt mix. The only exception shall be in accordance with the provisions of SS 515.05.07, where the Ministry shall have the ability to adjust the ratio of the components of the Hot-In-Place Recycled Pavement based upon the Contractor's Quality Control results and visual inspections.

The Ministry Representative may issue a Stop Work Order to the Contractor if the Contractor fails to adhere to the Quality Control Plan. The Ministry Representative will not issue a Resume Work Order until the Contractor has given the Ministry Representative satisfactory assurance that it has in place adequate capacity to fulfill the requirements of the Quality Control Plan.

515.04.02 Quality Control Plan – The Contractor shall prepare a detailed, written Quality Control Plan, based on the guidelines as shown in APPENDIX 515-A and functions integrally with any other Quality Management provisions of the Contract. The Contractor's Quality Control Plan shall be submitted to the Ministry Representative not less than five working days before starting the production of paving aggregate. The Ministry Representative will evaluate the Contractor's Quality Control Plan and will respond, in writing, within five days. Any subsequent changes to the Contractor's Quality Control Plan must also be submitted to the Ministry Representative for evaluation.

The Contractor's Quality Control Plan shall include a description and schedule of the intended sampling, testing and reporting. The Quality Control Plan must also include a detailed description of the means by which the Contractor shall use the quality control test results to ensure that the asphalt materials, aggregate, mix production, recycling and pavement compaction processes will be controlled to keep the end product within the specified limits. The Quality Control Plan must clearly show the flow of information from the quality control laboratory to the individuals who shall make the actual adjustments to the processes and equipment to this control. The plan will show time allowance for each step, the names and positions of all the people involved, and a clear description of the responsibilities of each.

515.04.03 Quality Control Testing and Inspection – The Contractor shall provide and maintain equipment and qualified personnel to perform all laboratory testing, field testing and inspection necessary to determine and monitor the characteristics and properties of all the materials produced and incorporated into the Work. They shall also monitor the workmanship of the final product in accordance with the Quality Control Plan as most recently submitted and accepted.

The Contractor shall use a qualified registered member of the Association of Professional Engineers and Geoscientists of British Columbia or a qualified registered member of the Applied Technologists and Technicians of British Columbia. This person shall be designated as the Quality Control Manager for the purposes of these specifications. The Quality Control Manager shall be responsible for preparation and sign off of the Quality Control Plan, shall be responsible for all Quality Control testing and inspections and shall be responsible for the signing of all Quality Control testing and inspection records and submissions to the Ministry.

The Contractor shall provide a testing facility(s) that meets the requirements necessary to carry out all the test procedures listed within this Section. The facility(s) must have the equipment specified under the appropriate test designation to perform the tests.

All equipment and the laboratory shall be well maintained and in good working condition. All testing equipment shall be calibrated, and evidence of the calibration shall be provided when requested by the Ministry Representative.

515.04.04 Quality Control Records – The results from quality control testing shall be reported on test logs and plotted on charts immediately after each test is completed. These test results, reports and charts shall be available within 24 hours of the end of each shift, to the Ministry Representative and at all times during the progress of the Work.

For the purposes of confirming delivery of asphalt mix to the road and the calculation of material application rates, the Contractor shall prepare and provide to the Ministry Representative, a copy of the Road Checker's Summary and the weigh tickets for each load received at the recycling operation at the end of each shift. The Road Checker's Summary shall include, but not be limited to, the following information:

(a) Truck Number

(b) Weigh Ticket Number and Net Weight of Load

(c) Time and location by station of delivery

(d) Admix and Rejuvenator Material Application Rate Calculations

(e) Processed Depth, Lengths, and Widths

(f) Temperatures (behind screed, windrow, ambient and admix)

(g) Notes pertaining to the paving of the appurtenances (intersections, tapers, etc.)

515.04.05 Final Quality Control Testing Reports – Prior to issuing a Completion Certificate, the Contractor shall provide the Ministry Representative with:

(a) A summary of all virgin aggregate quality control test results;

(b) Copies of all quality control test results for asphalt admix properties, Hot-In-Place Recycled Pavement and compaction; and

(c) Copies of all quality control charts.

515.04.06 Minimum Acceptable Construction Practices – Good construction practices shall be considered as standard construction procedures to be followed and shall include but not be limited to the procedures described below.

(a) **Recycling** – The existing highway pavement shall be heated, milled, reprocessed, with or without the addition of a rejuvenating agent or virgin mix as specified.

(b) **Finished Cross Fall Drainage** – Contractor shall ensure that the appropriate cross slopes for safety and positive drainage are maintained or improved after the recycling process.

(c) **Longitudinal Edges** – Contractor shall ensure that the longitudinal edges of the recycled pavement are blended to conform in elevation with the adjacent pavement unless this surface is scheduled to be recycled.

(d) **Excess Materials** – Excess materials shall be removed and deposited in a location such that they can be reincorporated into the recycled mixture. At no time shall excess material be cast across the newly processed surface or incorporated into the granular shoulders.

(e) **Compaction and Finishing** – The Contractor shall supply roller compaction behind the reprocessing operation to achieve densities and smoothness. The recycled mat shall have a uniform appearance and uniform surface. Any deficiencies shall be corrected at the Contractor's expense, to the satisfaction of the Ministry Representative.

(f) **Stockpiles** – Stockpiles of different types of material shall be located and constructed in such a manner as to prevent intermingling of the types and segregation of material.

(g) **Haul Vehicles** – Lubrication of the truck boxes with diesel fuel will not be permitted. All vehicles shall have adequately insulated truck boxes and shall be equipped with an insulating tarpaulin of such size as to completely protect the asphalt mix.

(h) **Existing Paved Surfaces** – Existing paved surfaces shall be clean to avoid contamination of the recycled pavement. The existing surfaces shall be swept back at least 20 cm wider than the area to be processed.

(i) **Existing Utility Structures** – The Contractor shall ensure all catch basins and manholes, which are within the operating area of the heating units, are clear of any flammable liquids or otherwise hazardous fumes/gases.

(j) **Heating of Recycled Material** – The recycled material shall not be heated over an average of 150°C in order to avoid excessive oxidation and hardening of the recycled Asphalt Cement.

(k) **Minimum Admix Temperatures** – Based on the most recent Temperature Viscosity Curve supplied by the Asphalt Cement Supplier. This information will be used to set minimum and maximum mixing temperatures.

(l) **Longitudinal Joints** – Longitudinal joints in the top lift of asphalt pavement will generally only be permitted where lane dividing lines are to be painted.

(m) **Transition Joints** – At the locations where recycling operations begin and end, the Contractor shall ensure transition joints between the processed and unprocessed surface are smooth and without irregularities. If any bumps or irregularities occur resulting from the reprocessing operation, the Contractor shall repair these areas with fresh asphalt mix and/or additional reprocessing, at the Contractor's own expense, as directed by the Ministry Representative.

(n) **PG Asphalt Cement** – Shall meet requirements set out in Special Provisions and SS 952.

(o) **Minimum Temperature** – Minimum Temperature behind the paver screed of the recycled material shall not be less than 110°C, unless the Ministry Representative accepts that the circumstances limit the reasonably achievable temperature to a lower value.

(p) **Rejuvenating Agent Application** –To accurately proportion the rejuvenating agent into the milled mixture the equipment shall comply with the following requirements:

(i) Positive feed and shut off of the rejuvenating agent, governed by the movement of the reprocessing machine;

(ii) Mechanically or electronically controlled application of the rejuvenating agent relative to and variable with the processing rate of the equipment;

(iii) A mechanical or electronic metering device capable of supplying an accurate indication of the quantity of rejuvenating agent being applied per square metre; and

(iv) A mechanical, electronic or manual method of measuring quantity of rejuvenating agent in the tank.

(q) Metering of Rejuvenating Agents – To accurately proportion the rejuvenating agent into the hot milled material, the equipment shall control the quantity of the rejuvenating agent to ± 0.05 L/m^2 of surfaced reprocessed with an agent application range of 0.0 L/m^2 to 1.0 L/m^2.

The measurement of the rejuvenating agent applied by means of a metering device shall be capable of recording accumulated litres to the accuracy of ± 2% and be fully visible to both the operator and the Ministry Representative.

Heating of Rejuvenating Agent shall be kept to within ± 5°C of the application temperature established by the manufacturer and/or the Ministry Representative.

(r) Outer Edges – The outer edge of each lane shall generally be tapered to a 12:1 slope.

(s) Recycling Adjoining Mats – When recycling an adjoining parallel mat there shall be at least a 50 mm overlap onto the previously recycled surface. This overlap shall be properly "raked" to form a homogeneous bond between the two mats.

(t) Raking – Any material that is raked from the joint shall not be placed on the new mat or placed on or in front of the paver. This material may be introduced at a point of the operation where the material shall be reheated and remixed with the Hot-In-Place Recycled Pavement.

(u) Discontinued Paving – At locations where reprocessing operations begin and end, the Contractor shall ensure that the transition between the processed and unprocessed surface is smooth and without irregularities. If any irregularities occur resulting from the reprocessing operation, the Contractor shall repair these areas with fresh asphalt mix and/or additional reprocessing, at the Contractor's own expense.

(v) Pavement Drainage – The Contractor shall at all times ensure that the appropriate cross slopes for safety and positive pavement drainage are maintained.

(w) Rollers – Rollers shall normally operate with the drive wheel nearest the paver and at a speed not in excess of 8 km/h. They shall not be allowed to park on the mat prior to complete cooling.

515.05 Supply of Virgin Aggregates, Asphalt Materials, Asphalt Admix Design and Job Mix Formula

515.05.01 Work in Ministry Pits or Quarries – When operating in a Ministry pit or quarry, the Contractor shall comply with all provisions of SS 145.26.

515.05.02 Supply of Virgin Aggregates, Aggregate Production and Characteristics – The Contractor shall not produce paving aggregate until the Contractor has received written notification that the Quality Control Plan is in accordance with SS 515.04.02, and has in place testing facilities for aggregate production that are in accordance with the Quality Control Plan.

For the production of virgin Asphalt Mix Aggregate, within Ministry pits, the Contractor shall provide crushing equipment that will permit all aggregate, that passes through 375 mm x 450 mm slotted openings, to be used for the production of crushed aggregate. Rocks, which will not pass through these openings, shall be stockpiled or disposed of as directed by the Ministry Representative. Crushing and screening equipment shall be provided with adequate facilities and capacity to be able to bleed off reject aggregate and remove any excess fine aggregate, dust or objectionable aggregate coatings, to make it acceptable for use. All products from crushing or screening plants, that can be used shall not be wasted, and shall be stockpiled or used as directed by the Ministry Representative.

Where the Ministry has available any test results information on the properties shown in Table 515-B, for a Ministry pit or other source, the Ministry Representative will upon request, provide that information to the Contractor. Otherwise, sampling and testing to determine and demonstrate the compliance of paving aggregate with the requirements of this Section shall be the responsibility of the Contractor.

Paving aggregate shall meet the requirements of Table 515-D for medium mix asphalt aggregates, be within the variations limits specified in APPENDIX 515-B, Table 515-J, as well as the following requirements:

(a) Virgin Coarse Aggregate:

(i) Shall be all mineral matter retained on the sieve designated in the test procedures for each individual test.

(ii) Shall consist of crushed stone, crushed gravel, or combination thereof, or materials naturally occurring in a fractured condition, or materials naturally occurring of highly angular nature or rough texture.

(iii) Shall be free from coating of clay, silt or other deleterious material, and shall meet the requirements listed in Table 515-B and Table 515-F.